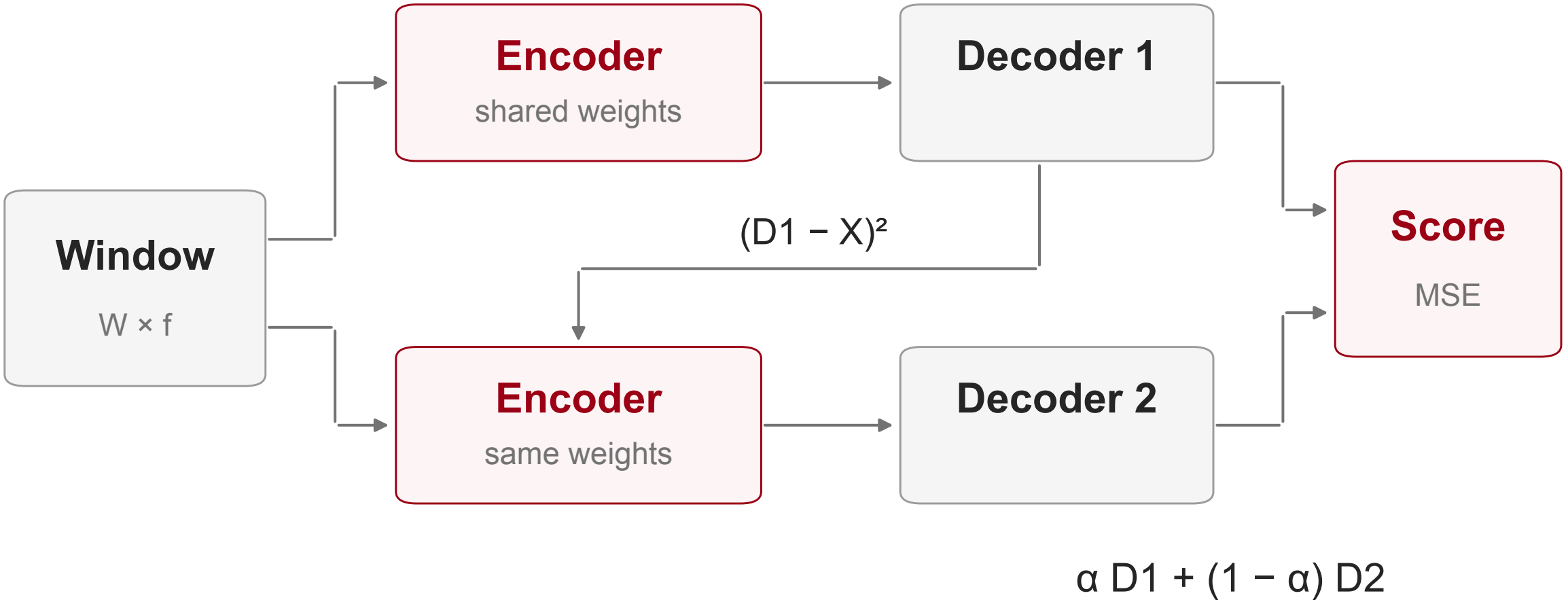
TranAD+


Window
W × f
Encoder
shared weights
Decoder 1
(D1 − X)²
Encoder
same weights
Decoder 2
Score
MSE
α D1 + (1 − α) D2

Datasets & study

Dataset	Mission	Inputs	Labels	W0	FF0
OPS-SAT	ESA CubeSat	1	Segments	10	8
SMAP	Earth observation	25	Time points	10	50
MSL	Mars rover	55	Time points	50	110

30
configurations / dataset

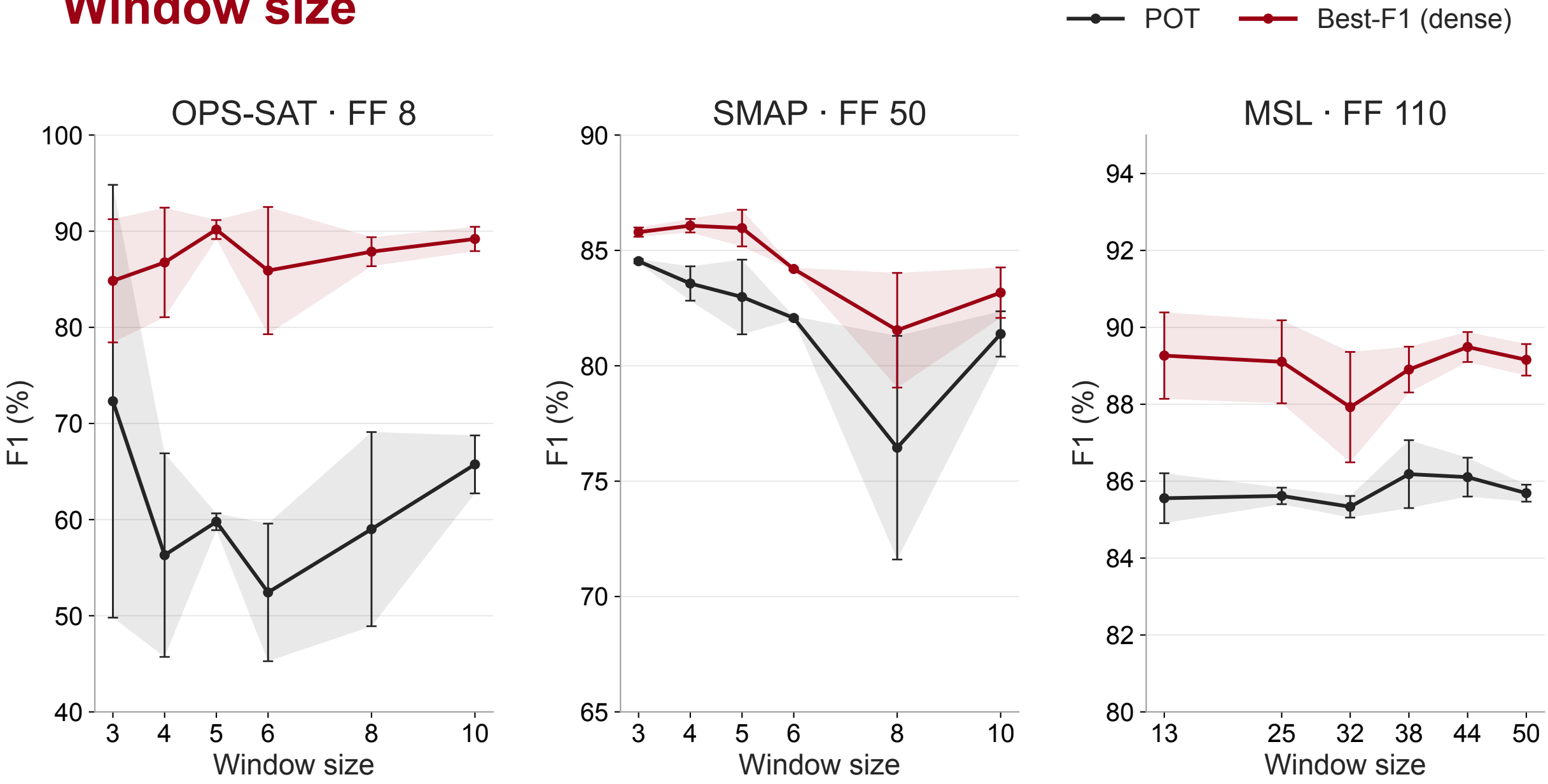
3
seeds

50
epochs

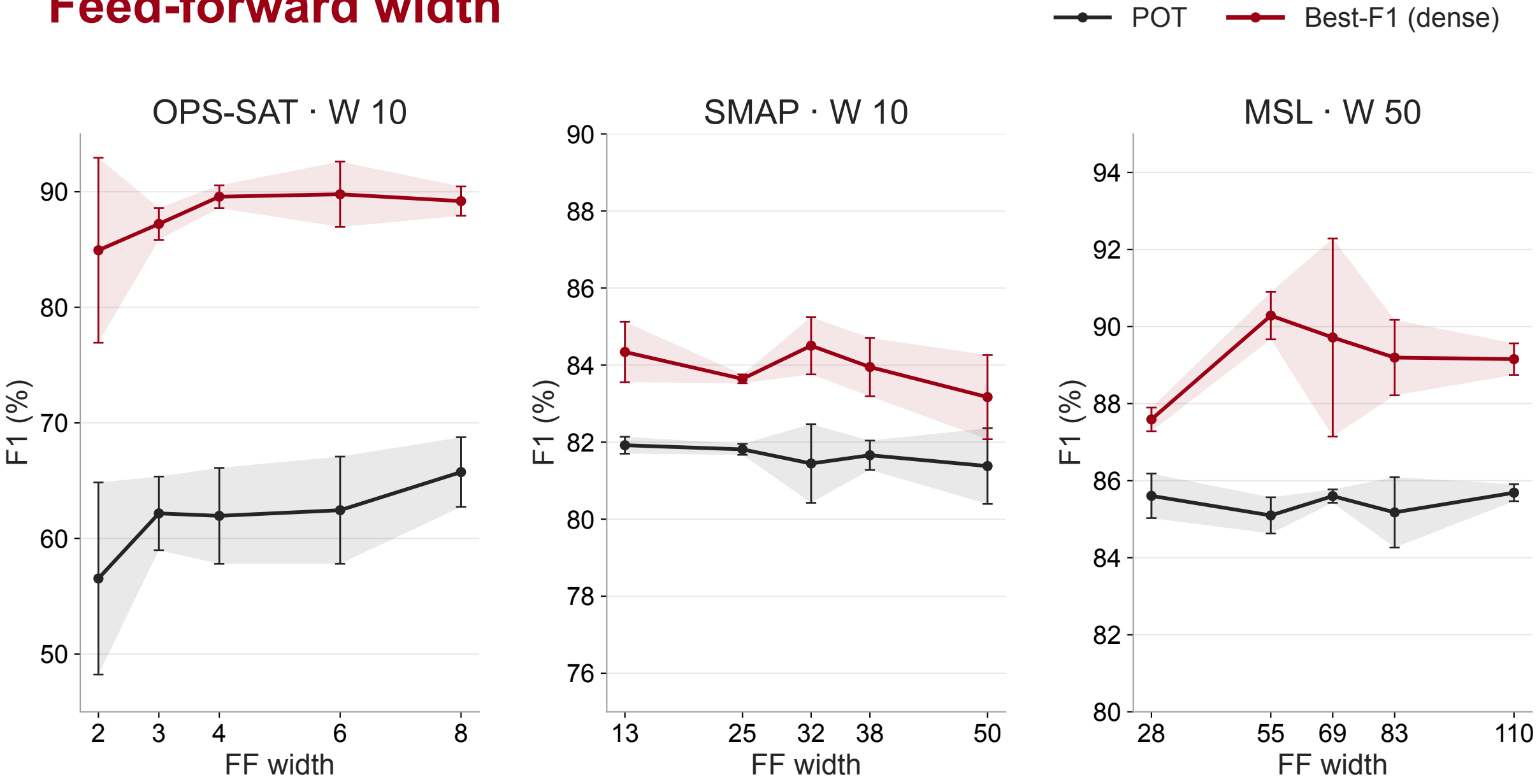
Benchmarks · F1 (%)

Dataset	Paper POT	Our POT	Paper Best-F1	Our Best-F1
DSN 1k	92.47	92.34	80.44	80.73
SMAP	84.24	84.22	84.91	84.70
MSL	89.01	90.17	81.96	79.03
SWaT	32.90	32.49	82.39	82.07
SMD	71.29	67.04	85.09	84.06
WADI	50.84	17.74	27.88	27.88

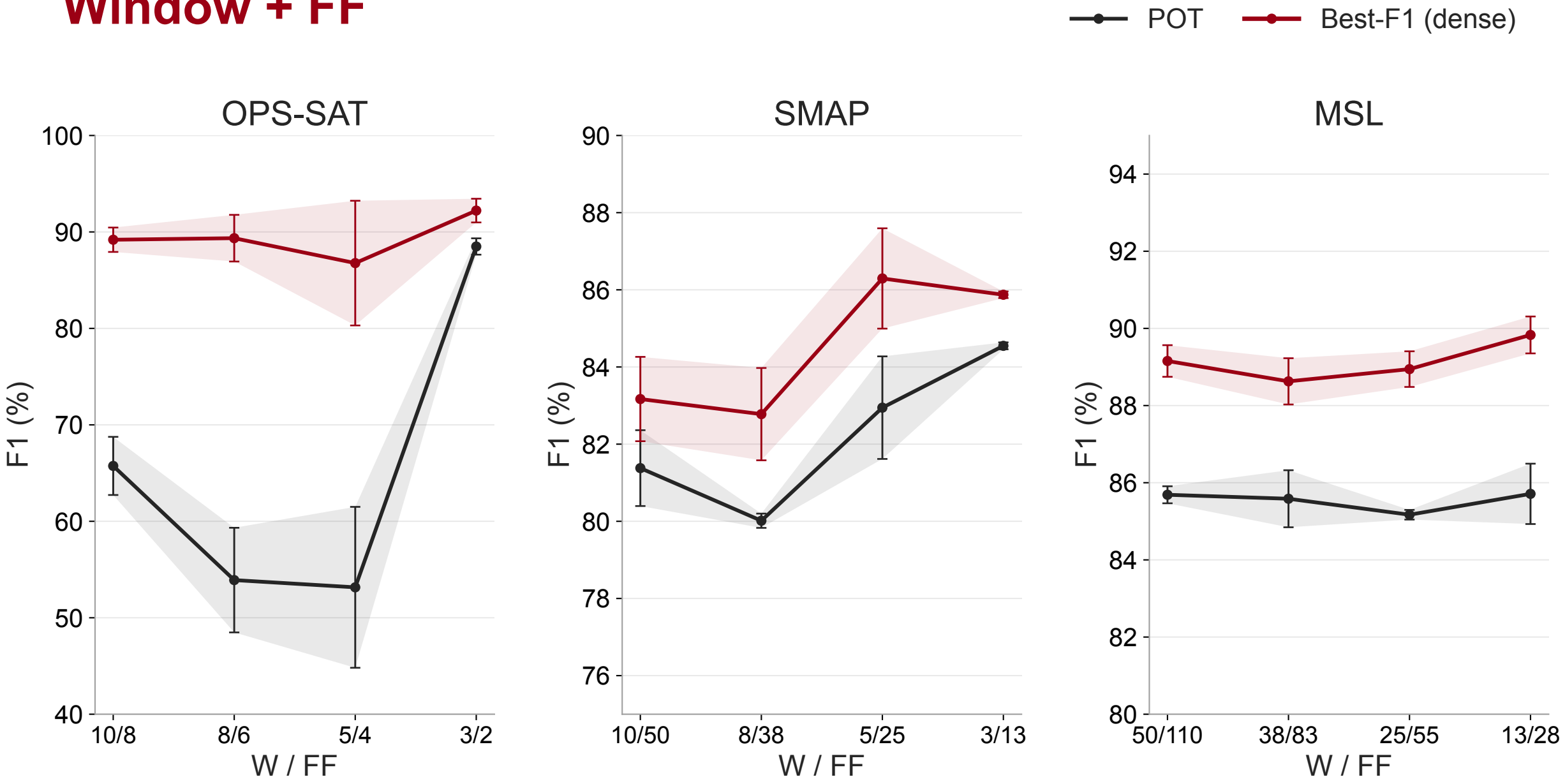
Window size


POT
Best-F1 (dense)
OPS-SAT · FF 8
SMAP · FF 50
MSL · FF 110
F1 (%)
Window size
100
90
80
70
60
50
40
85
75
65
94
92
88
86
84
82
3
4
5
6
8
10
13
25
32
38
44
50

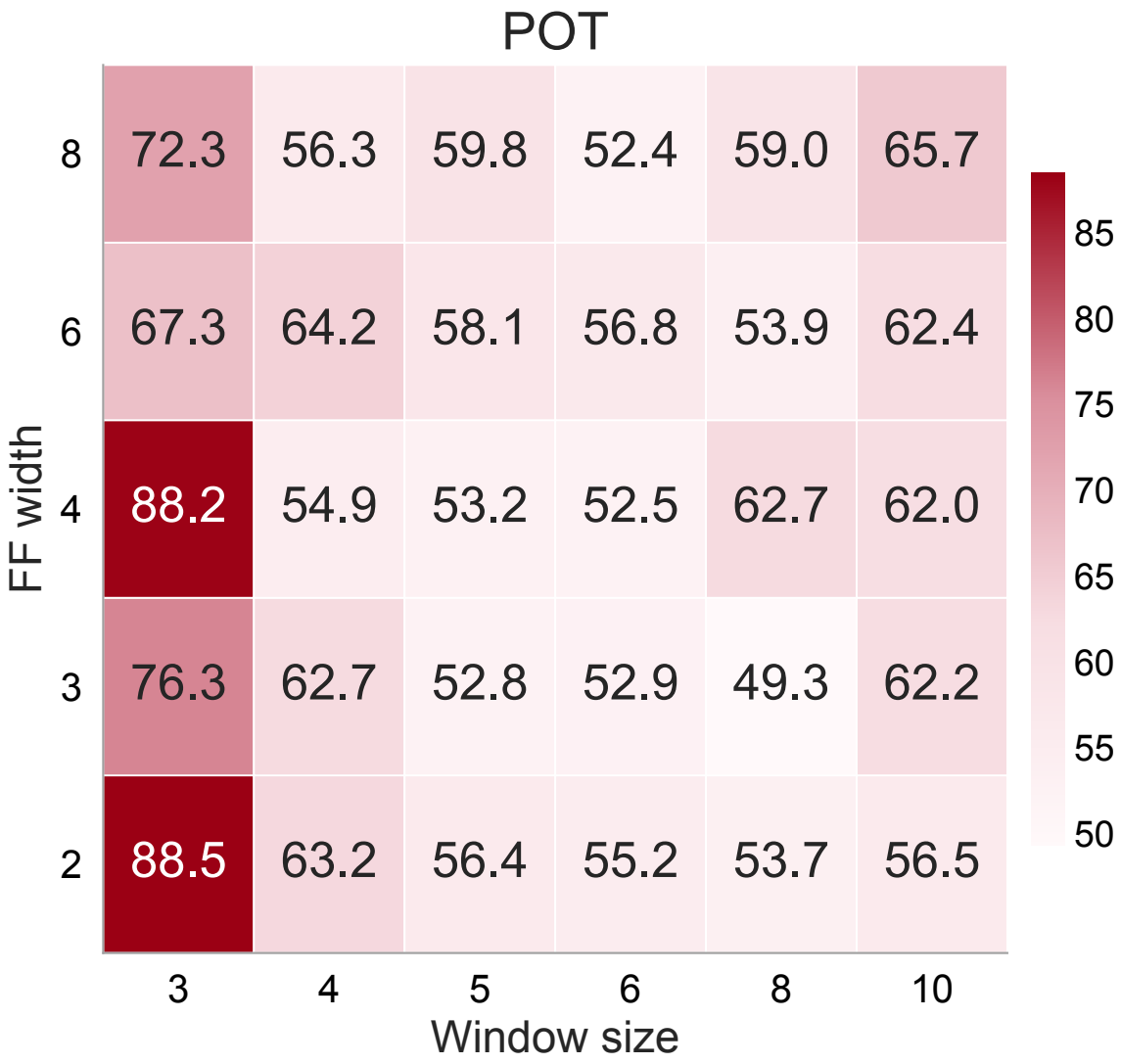
Feed-forward width

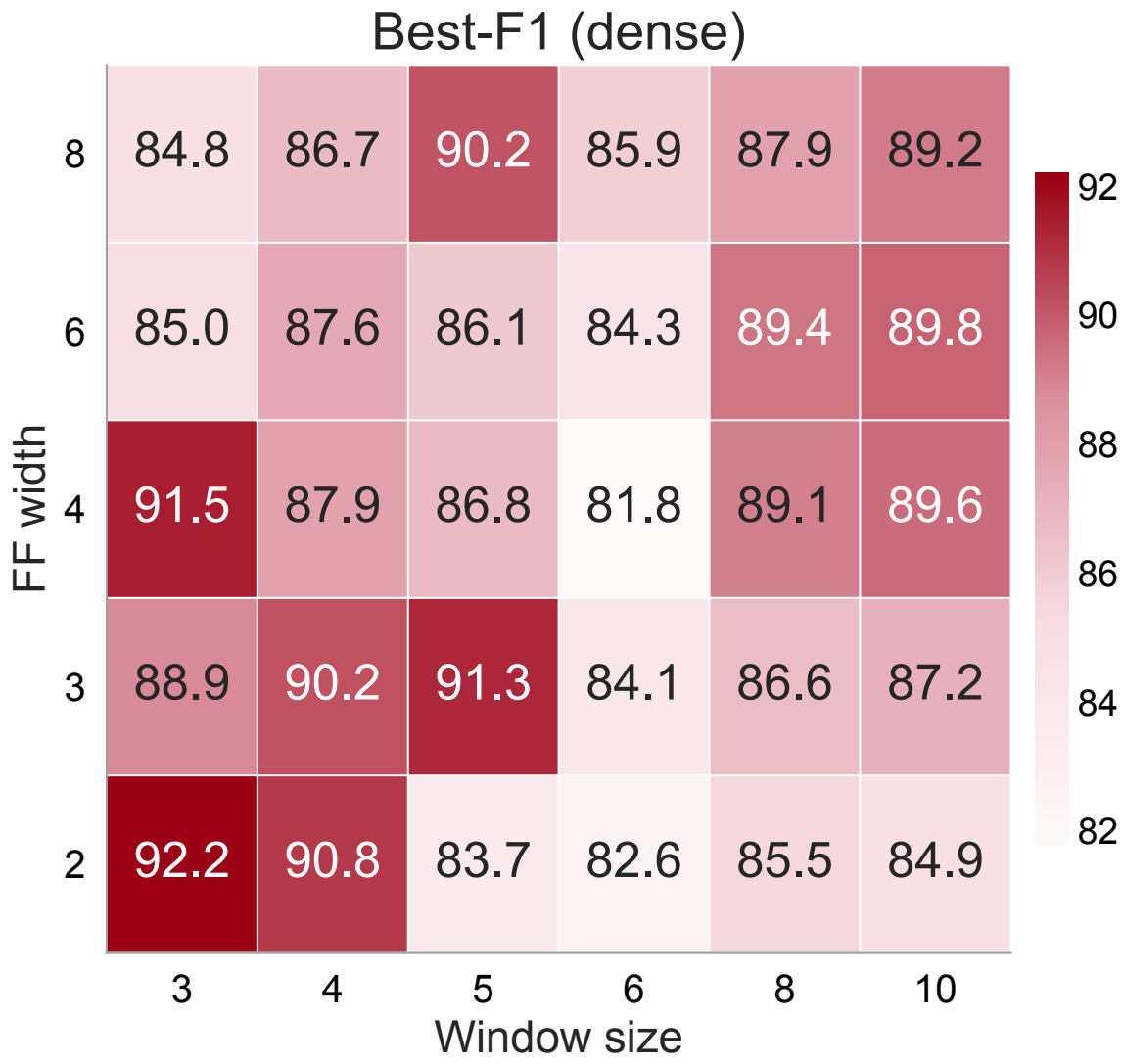


Window + FF

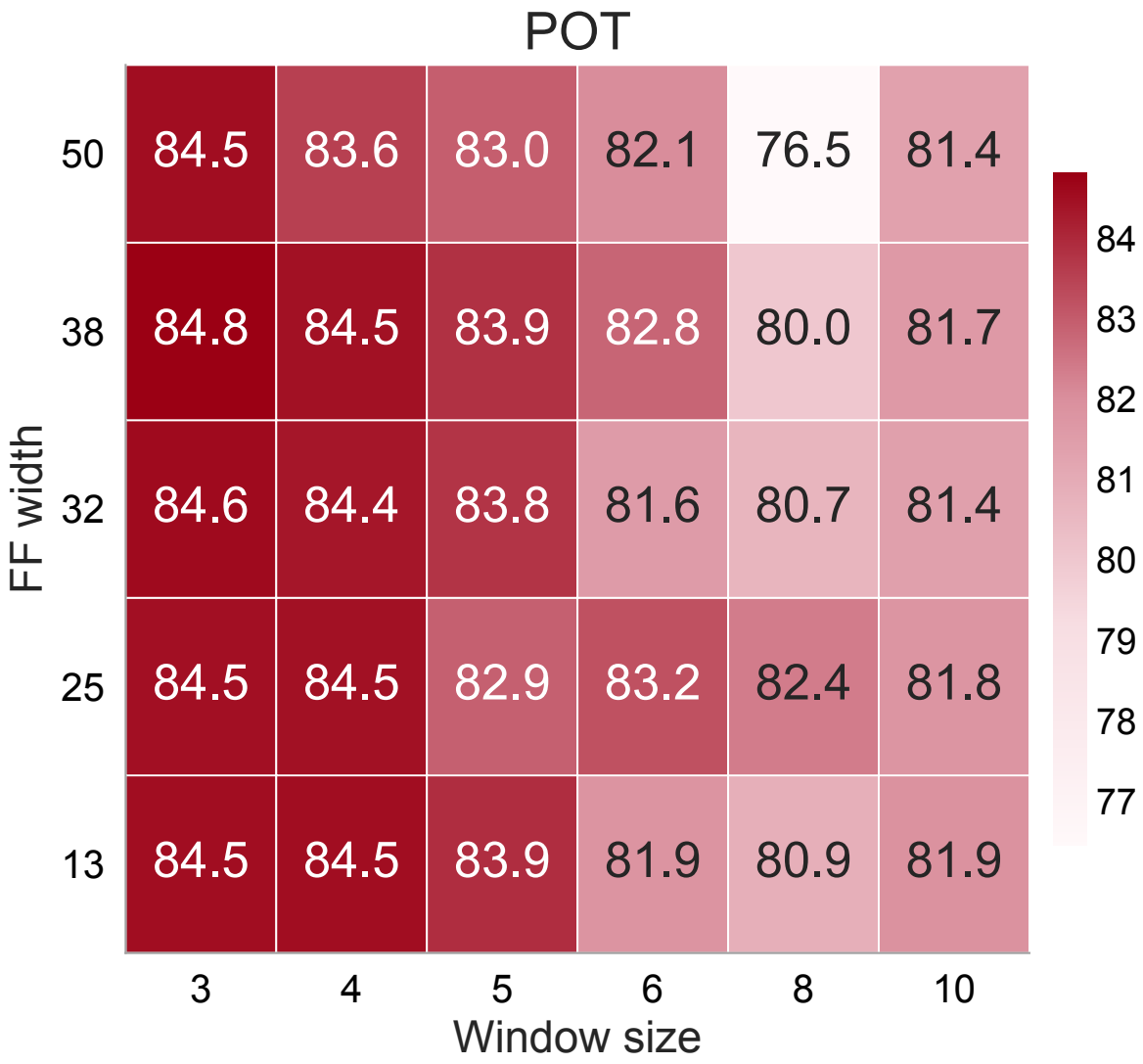

POT
Best-F1 (dense)
OPS-SAT
SMAP
MSL
F1 (%)
W / FF
10/8
8/6
5/4
3/2
10/50
8/38
5/25
3/13
50/110
38/83
25/55
13/28

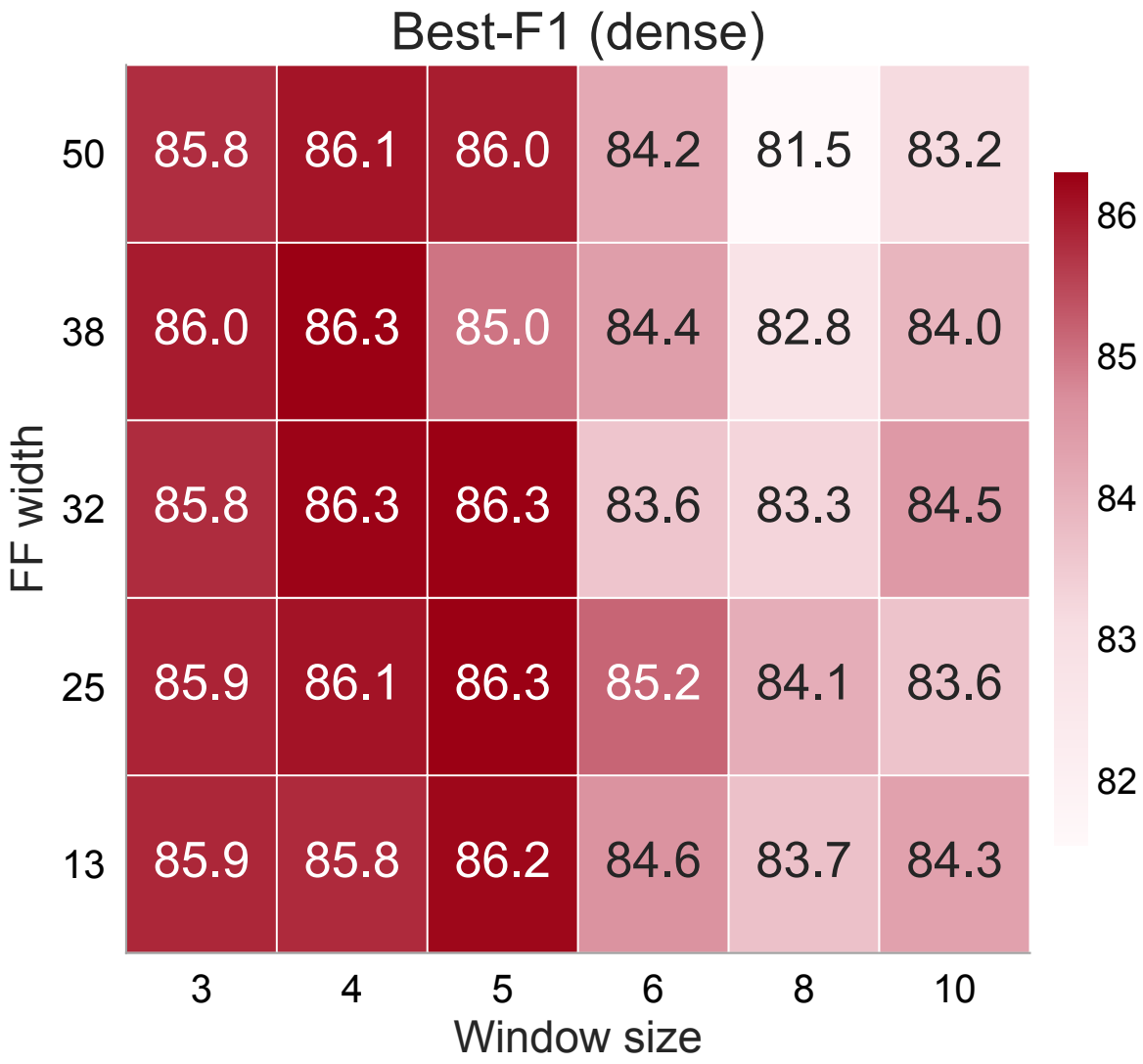
OPS-SAT


POT
FF width
8
6
4
3
2
72.3 56.3 59.8 52.4 59.0 65.7
67.3 64.2 58.1 56.8 53.9 62.4
88.2 54.9 53.2 52.5 62.7 62.0
76.3 62.7 52.8 52.9 49.3 62.2
88.5 63.2 56.4 55.2 53.7 56.5
3 4 5 6 8 10
Window size
85
80
75
70
65
60
55
50

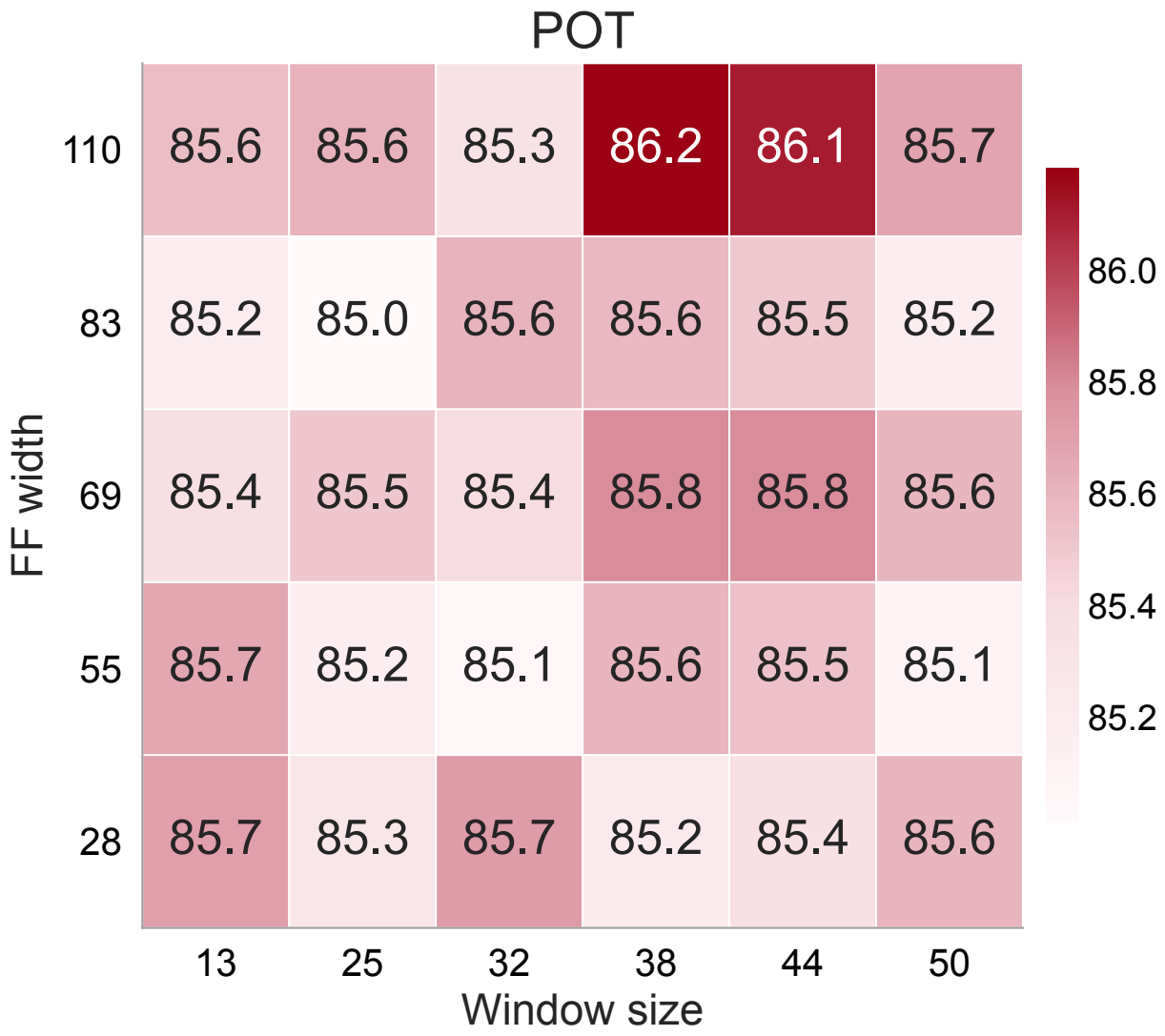

Best-F1 (dense)
FF width
8
6
4
3
2
84.8 86.7 90.2 85.9 87.9 89.2
85.0 87.6 86.1 84.3 89.4 89.8
91.5 87.9 86.8 81.8 89.1 89.6
88.9 90.2 91.3 84.1 86.6 87.2
92.2 90.8 83.7 82.6 85.5 84.9
3 4 5 6 8 10
Window size
92
90
88
86
84
82

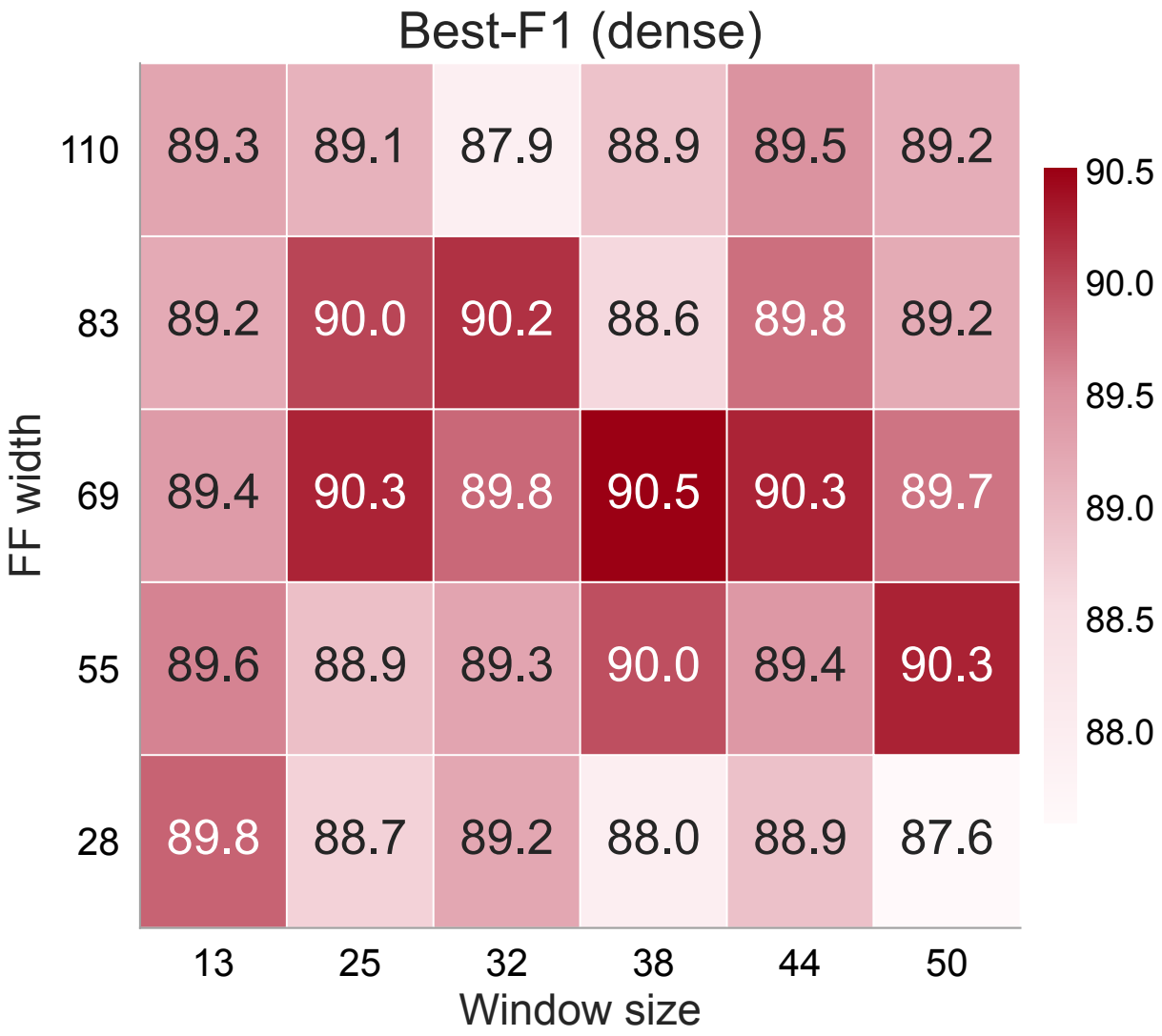
SMAP


POT
FF width
50
38
32
25
13
84.5 83.6 83.0 82.1 76.5 81.4
84.8 84.5 83.9 82.8 80.0 81.7
84.6 84.4 83.8 81.6 80.7 81.4
84.5 84.5 82.9 83.2 82.4 81.8
84.5 84.5 83.9 81.9 80.9 81.9
3 4 5 6 8 10
Window size
84 83 82 81 80 79 78 77

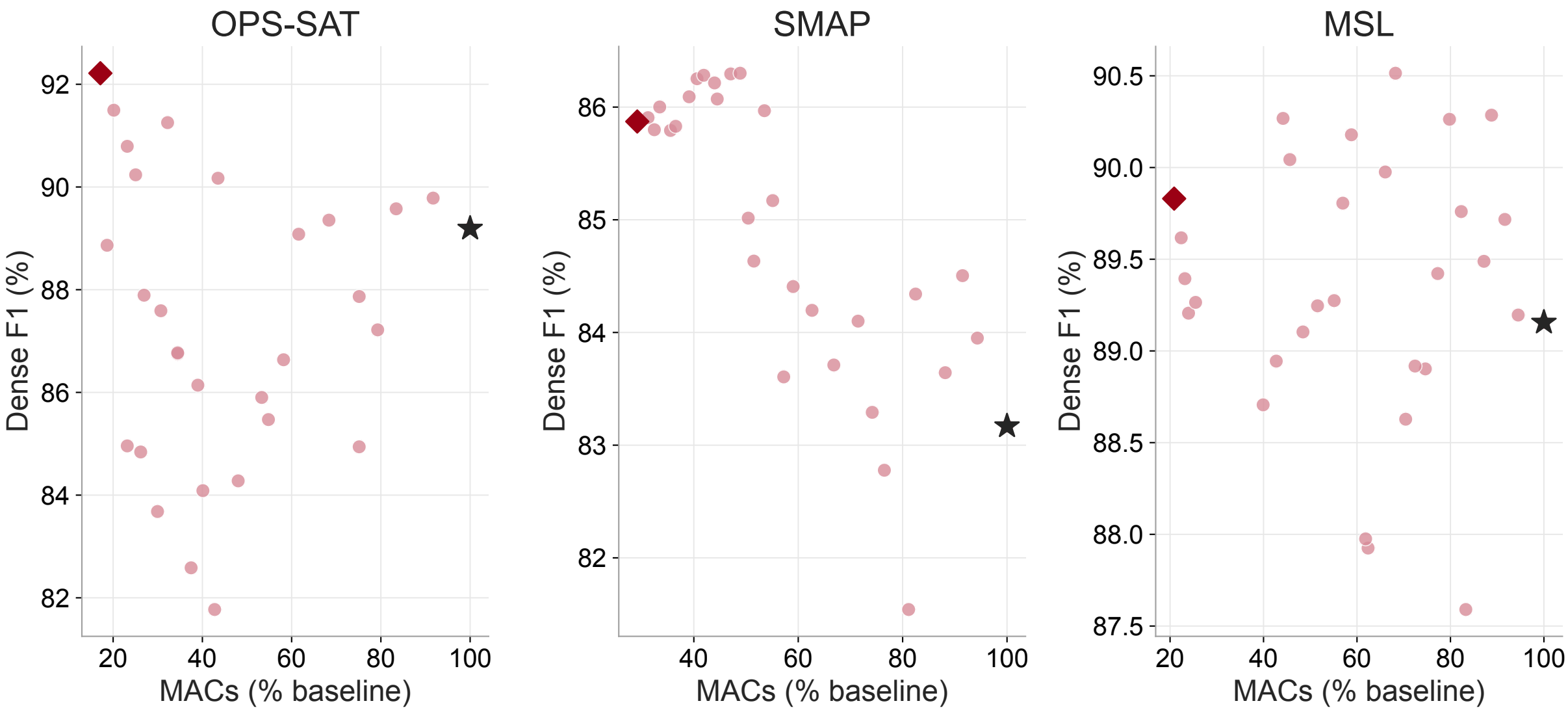

Best-F1 (dense)
FF width
50
38
32
25
13
85.8 86.1 86.0 84.2 81.5 83.2
86.0 86.3 85.0 84.4 82.8 84.0
85.8 86.3 86.3 83.6 83.3 84.5
85.9 86.1 86.3 85.2 84.1 83.6
85.9 85.8 86.2 84.6 83.7 84.3
3 4 5 6 8 10
Window size
86 85 84 83 82

MSL


POT
FF width
110
83
69
55
28
85.6 85.6 85.3 86.2 86.1 85.7
85.2 85.0 85.6 85.6 85.5 85.2
85.4 85.5 85.4 85.8 85.8 85.6
85.7 85.2 85.1 85.6 85.5 85.1
85.7 85.3 85.7 85.2 85.4 85.6
13 25 32 38 44 50
Window size
86.0
85.8
85.6
85.4
85.2


Best-F1 (dense)
FF width
110
83
69
55
28
89.3 89.1 87.9 88.9 89.5 89.2
89.2 90.0 90.2 88.6 89.8 89.2
89.4 90.3 89.8 90.5 90.3 89.7
89.6 88.9 89.3 90.0 89.4 90.3
89.8 88.7 89.2 88.0 88.9 87.6
13 25 32 38 44 50
Window size
90.5
90.0
89.5
89.0
88.5
88.0

Accuracy vs compute



Compact models

Dataset	Model	W	FF	Parameters	MACs	Dense F1
OPS-SAT	Baseline	10	8	249	2,124	89.19
	Compact	3	2	159	364	92.22
SMAP	Baseline	10	50	67,575	464,700	83.17
	Compact	3	13	56,364	135,500	85.87
MSL	Baseline	50	110	323,565	11,008,140	89.16
	Compact	13	28	269,199	2,297,900	89.83

Precision · Recall · AUC

Dataset	Model	Dense F1	Precision (%)	Recall (%)	PA AUC (%)
OPS-SAT	Baseline	89.19	86.90	91.73	91.82
	Compact	92.22	92.74	91.81	93.80
SMAP	Baseline	83.17	85.53	81.16	89.56
	Compact	85.87	91.98	80.52	89.75
MSL	Baseline	89.16	87.44	90.95	94.72
	Compact	89.83	88.25	91.48	95.02

Architecture proposals

- **Shared decoder**

  One set of decoder weights · two passes

- **Single pass**

  One encoder–decoder pass · no feedback

- **Encoder only**

  Encoder + reconstruction head